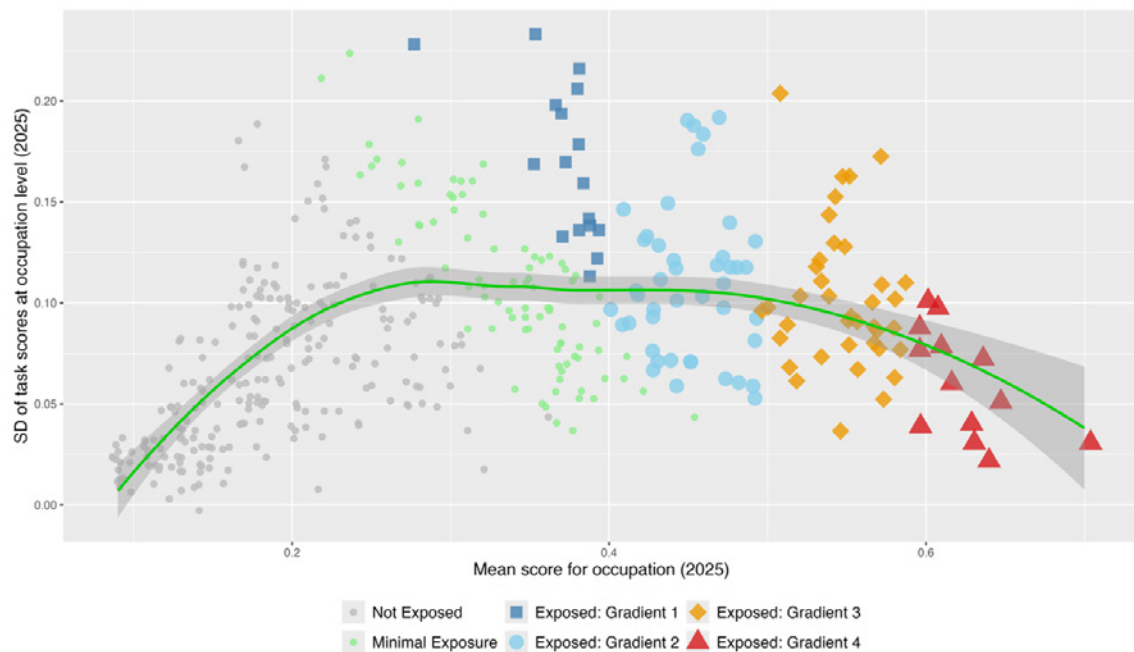


► 8 Adjusted Global Index of GenAI Exposure

To classify ISCO-08 occupations into varying levels of exposure to Generative AI (GenAI), we update the framework introduced in Gmyrek et al. (2023). Similarly to the original approach, we rely on two moments of distribution: occupational mean and standard deviation (SD) of task-level scores pertaining to a given 4-digit occupation. However, instead of using automation and augmentation potential as two extreme categories with the "big unknown" in between, we now adopt a more nuanced classification scheme that represents a spectrum of exposure, with occupations considered as exposed classified into four increasing gradients (Figure 16 and Table 5).

► **Figure 16. AI Exposure Gradients: ISCO-08 4-digit level occupations**



Note: Figure 16 presents all ISCO-08 4-digit occupations classified into GenAI exposure categories as per the criteria in Table 5. A detailed list of occupation-level means and SDs is available in the Appendix, Table A1. In addition, task-level scores within the overall hierarchy of ISCO-08 are available here: https://pgmyrek.github.io/2025_GenAI_scores_ISCO08/

Gradient 1 represents occupations with low overall GenAI exposure but significant variability across tasks (Low exposure, high task variability). While some tasks within these roles may have high automation potential, the occupation has many tasks that continue requiring human roles, aligning closely with the notion of augmentation in the old framework. Gradient 2 includes occupations with moderate GenAI exposure and a mix of highly exposed and minimally exposed tasks, resulting in uneven impacts where some tasks may be disrupted while others remain unaffected (Moderate exposure, high task variability). Gradient 3 captures occupations where a significant portion of tasks are consistently exposed to GenAI, signaling growing automation risks and requiring adaptation strategies for workers (Significant exposure, high task variability). Gradient 4 highlights occupations with the highest share of tasks exposed to potential GenAI-driven automation, and with a high consistency of this exposure across tasks (Highest exposure, low task variability). This gradient aligns closely with the notion of automation risk in our earlier classification system (Gmyrek et al., 2023). In addition, among occupations outside these four categories, we introduce a more granular distinction of Minimal Exposure, where some interaction with GenAI

may occur, but without significantly altering the nature of these roles, and the lowest category of Not Exposed occupations, where GenAI currently has no observable potential to automate tasks.

► **Table 5. GenAI Exposure Gradients: Definition and Interpretation**

GenAI Exposure	Definition (Mean & SD of Task Scores)	Interpretation
Exposed: Gradient 4 (Highest exposure, low task variability)	$\mu \geq 0.6$ and $\mu - \sigma \geq 0.5$	High and consistent GenAI exposure across tasks within the occupation. Most current tasks in these jobs have a high potential of automation, with little variability in task-level exposure.
Exposed: Gradient 3 (Significant exposure, high task variability)	$0.5 \leq \mu < 0.6$ and $\mu + \sigma \geq 0.5$	Above-moderate occupational exposure: even though some tasks remain less exposed, the overall potential of automation of the current tasks with GenAI is growing in these occupations.
Exposed: Gradient 2 (Moderate exposure, high task variability)	$0.4 \leq \mu < 0.5$ and $\mu + \sigma \geq 0.5$	Moderate occupational AI exposure, with high task-level variability. These occupations include a mix of some tasks that are exposed to GenAI and others not at risk, making the impact uneven.
Exposed: Gradient 1 (Low exposure, high task variability)	$\mu < 0.4$ and $\mu + \sigma \geq 0.5$	Low overall GenAI exposure at the occupational level, but high variability across tasks. Some tasks within these occupations have an elevated automation potential, even if the occupation as a whole remains strongly reliant on tasks that have a low potential of automation.
Minimal Exposure (Low exposure, moderate task variability)	$\mu < 0.5$ and $\mu + \sigma > 0.4$	Occupations with low GenAI exposure, where some tasks show moderate automation potential, but overall occupational exposure remains limited.
Not Exposed	Occupations that don't meet any of the above conditions.	Occupations where most tasks remain relatively unaffected by GenAI, with low task variability and a stable low mean exposure score.

This revised framework addresses some limitations of the earlier categorization system, particularly the interpretative challenges posed by the category, “the big unknown”. By expanding the exposure categories into a continuous gradient, the updated framework improves the readability and interpretability of results and enables us to stack employment estimates into the overall category of exposed occupations (Figure 17 in Section 10).¹⁸ It also captures in a more nuanced way how GenAI can impact occupations at varying levels of exposure, based on task-level variability and overall occupational risks. Furthermore, the framework's flexibility enhances its applicability to country-specific contexts, allowing policymakers to better understand the distribution of GenAI's potential impact and prioritize interventions for the most affected groups in the national contexts.

¹⁸ One of the limitations of the current approach is that neither ISCO-08 nor the Polish classification system includes information on the relative frequency or importance of tasks within occupations. As a result, all tasks are weighted equally when calculating exposure scores. Future research could refine this by incorporating task-level weights, for instance by mapping to occupational databases such as O*NET or by collecting self-reported data on task relevance directly from workers. A similar refinement could concern mapping the level of skills to individual tasks.

disparities are even more pronounced in High-Income Countries, where 34% of total employment falls within one of the four exposure gradients, with 17.3% in gradients 3 and 4. However, when disaggregated by gender, 28% of male employment is exposed, with only 3.5% in gradient 4, compared to 41% of female employment, with 14.4% in gradient 3 and 9.6% in gradient 4.

Income-based differences in exposure are striking, with High-Income Countries showing the highest share of employment in one of the four exposure gradients (34%). The total share of exposed employment declines significantly as income levels decrease, reaching just 11% in Low-Income Countries (LICs). Lower income levels also reduce gender disparities in exposure, primarily due to the lower concentration of occupations in the highest exposure gradients (gradients 3 and 4). Gender disparities are also more pronounced in wealthier regions, particularly in Europe and Central Asia, where 39% of female employment falls into one of the four exposure gradients compared to 26% for men. These patterns reflect both occupational structures and the extent to which GenAI-exposed roles, such as clerical, financial, and customer service jobs, are concentrated in higher-income countries.

These new estimates are consistent with those given in 2023, but more refined. The previous version attributed 2.3% of global employment to the “automation” category, one percentage point lower than the 3.3% assigned to gradient 4. In addition, the 2023 estimate attributed 13% of employment as subject to “augmentation” and an additional 16.2% of employment in “the big unknown”. The combined employment share for Gradients 1-3 for the world in the current analysis is 20.5%, 9 percentage points lower than combined shares of augmentation and “the big unknown” (29.2%) reported in 2023. The 2023 findings that women are more affected than men, and that the percentage of employment affected increases as countries’ income level increases, still hold.

It is important to stress that these new estimates reflect “exposure” to GenAI and not the actual impact on employment. As in 2023, the 2025 estimates represent the upper threshold of the percentage of employment that could be affected if GenAI technology were fully implemented. Infrastructure constraints (electricity, broadband), insufficient digital skills, the cost of technology, and inherent operational difficulties associated with the technology are just some of the barriers to full adoption. An important aspect also concerns the institutional and societal acceptance of automating tasks and occupations – even when technically feasible. For instance, few consumers may be inclined to watch evening news presented by avatars. Similarly, financial institutions might hesitate to fully delegate decisions such as mortgage approvals or business plan evaluations to AI systems, preferring to retain human judgement, oversight and accountability.

Indeed, the results indicate that few jobs consist of tasks that are fully automatable with current GenAI technology; nearly all occupations have some tasks that require human input. Gradients 1-3 comprise occupations where the automation of specific tasks is limited, and some degree of task variability exists. As such, the “bundle of tasks” that constitutes occupations acts as a buffer against full automation. However, this does not mean that demand for these occupations will remain stable, as the efficiencies gained from introducing technology may mean that fewer workers are needed. If efforts are not made to help such jobs evolve with GenAI and integrate new tasks, even partial automation of existing tasks could lead to a decline in overall job demand in occupations found in higher exposure gradients. In addition, in some cases, the remaining set of non-automated tasks may result in jobs that are no longer humanly sustainable or desirable, thereby limiting the feasibility of adopting imbalanced technological solutions (Tegtmeier et al., 2022).